



(Year 2 / A2 Level)

Capacitors Lecture Notes

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CHAPTER 19

Capacitors

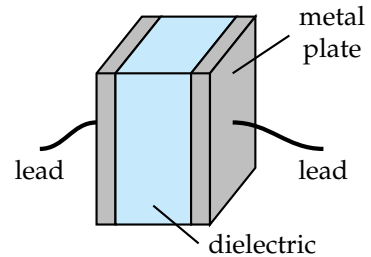
19.1 capacitors: an introduction

19.1.1 capacitors

capacitors are elementary electrical units widely used in electrical and electronic engineering

a typical capacitor has two conductive plates

between the plates there is usually an insulating material called *dielectric*



circuit symbol for a capacitor is $\text{---}||\text{---}$

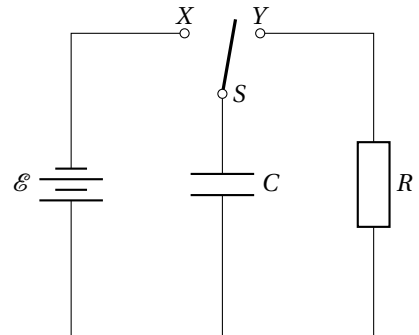
if we construct an electric circuit as shown

when contact S is moved to X , capacitor is connected to a voltage supply and becomes charged

positive and negative charges are separated onto two plates, and they will stay where they are even if we disconnect the capacitor from the voltage supply

if we then move S to Y , the charged capacitor discharges and drives a current through resistor R , i.e., it can act as a temporary power source^[1]

so capacitors can be used to store and release energy^[2]



^[1]Details on charging and discharging processes will be gone through in §19.4.

^[2]Other important functions of capacitors in electronic circuits include smoothing output voltage of power supplies, blocking direct current while allowing alternating current to pass, etc.

19.1.2 mutual capacitance

to describe ability of a capacitor to store charges, we define the notion of capacitance

(mutual) capacitance of a parallel-plate capacitor is defined as the ratio of the charge stored on one plate to the potential difference across the two plates.

in a word equation: mutual capacitance $C = \frac{\text{charge on one plate } Q}{\text{p.d. } V \text{ across the plates}} \Rightarrow C = \frac{Q}{V}$

➤ unit of capacitance: **farad** [3] : $[C] = F$

farad a derived unit: $1 F = 1 C \cdot V^{-1}$

farad is a large unit, more common subunits of capacitance in use are sub-multiples of farad:

$$1 \mu F = 10^{-6} F, \quad 1 nF = 10^{-9} F, \quad 1 pF = 10^{-12} F$$

➤ for a parallel-plate capacitor, charges on the two plates are equal but opposite

net charge on the capacitor: $Q_{\text{net}} = (+Q) + (-Q) = 0$

so we should emphasise on the notion of charge on *one* plate in the definition

➤ capacitance depends on *geometry* of the device and permittivity of the dielectric material

capacitance does not depend on electric field or potential^[4]

for example, capacitance between two metal plates is: $C = \frac{\epsilon_0 A}{d}$ ^[5]

A is area of plate, d is distance between plates, both are geometrical quantities

19.1.3 self-capacitance

there are two closely related notions of capacitance: *mutual* capacitance and *self* capacitance

the definition for capacitance given in the previous section, is actually *mutual* capacitance

on the other hand, all bodies are able to store electrical charge

^[3]The unit is named after Michael Faraday, a British physicist who developed the concept of capacitance. Faraday's other main discoveries include electromagnetic induction and electrolysis. He established the basis for the concept of the electromagnetic field in physics.

^[4]Recall the resistance of an electrical component. Resistance is defined as the ratio of p.d. to current, but the value of resistance is essentially dependent on the length, cross-sectional area and material of the component, instead of the p.d. applied or the current flowing through it.

^[5]If there is *dielectric* in between, the formula should be rewritten as $C = \frac{\epsilon A}{d}$, where ϵ is permittivity of dielectric. These formulae are not examinable by the syllabus.

any object that can be electrically charged exhibits capacitance

we define **self-capacitance** of an object as the amount of charge that must be added to increase per unit electrical potential

in a word equation, self capacitance $C = \frac{\text{charge of object } Q}{\text{electric potential of object } V} \Rightarrow C = \frac{Q}{V}$ [6]

Example 19.1 Self-capacitance of a charged metal sphere in a vacuum

consider a metal sphere of radius R and carries an electric charge of Q

its electric potential: $V = \frac{Q}{4\pi\epsilon_0 R}$

self-capacitance of the sphere: $C_{\text{sphere}} = \frac{Q}{V} = Q \times \frac{4\pi\epsilon_0 R}{Q} \Rightarrow C_{\text{sphere}} = 4\pi\epsilon_0 R$

note that capacitance is only dependent on its geometrical property (radius R) □

Example 19.2 A conducting sphere of radius 1.0 m is situated in free space. (a) Find its capacitance. (b) In order to raise its potential to 5000 V, find the amount of charge needed.

✍ capacitance of the sphere: $C = 4\pi\epsilon_0 R = 4\pi \times 8.85 \times 10^{-12} \times 1.0 \approx 1.11 \times 10^{-10} \text{ F}$

(here you can see farad being an impractically huge unit)

charge on sphere: $Q = CV = 1.11 \times 10^{-10} \times 5000 \approx 5.56 \times 10^{-7} \text{ C}$ □

analogy with ideal gases

an interesting analogy can be made between capacitors and ideal gases

recall an ideal gas is described by equation $pV = nRT$ [7]

compare $\left\{ \begin{array}{l} \text{amount of charge: } Q = CV \\ \text{amount of substance: } n = \left(\frac{V}{RT} \right) p \end{array} \right.$

volume V of a container has a certain space *capacity*, at fixed T , pumping more gas (increase n) into system, pressure p increases

similarly, a capacitor has a certain charge capacity, adding more charge Q increases p.d. V , so the quantity C is naturally called *capacitance*

also for a container, there exists a maximum pressure which it can withstand

[6] For either mutual capacitance or self capacitance, defining equation $C = \frac{Q}{V}$ takes the same form, but you should keep in mind that Q and V represent different things in different contexts.

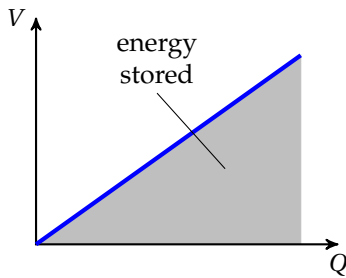
[7] Don't confuse voltage V with volume V !

for a capacitor, there exists a *breakdown voltage*, or *withstand voltage*, beyond which there could be sparking across the capacitor

19.2 energy stored in a capacitor

to charge a capacitor, need to push electrons off one plate and onto the other

separation of positive and negative charges requires work done $\Rightarrow$ energy is stored



since charge Q varies with p.d. V , we shall use the

V - Q graph to find work done W

area under V - Q graph is equal to work done W

$$W = \frac{1}{2} QV$$

substitute $Q = CV$, energy stored in capacitor is^[8]:

$$W = \frac{1}{2} CV^2 = \frac{Q^2}{2C}$$

Example 19.3 When the p.d. across a capacitor of 1.8×10^{-4} F is increased from 10 V to 20 V, how much additional energy is stored?

 energy change: $\Delta W = W_f - W_i = \frac{1}{2} CV_f^2 - \frac{1}{2} CV_i^2 = \frac{1}{2} \times 1.8 \times 10^{-4} \times (20^2 - 10^2) = 2.7 \times 10^{-2}$ J $\square$

Question 19.1 A capacitor of 2500 μ F is charged to a working voltage of 18 V. (a) What is the magnitude of positive charge on its plate? (b) What is the energy stored?

Question 19.2 A capacitor initially charged to a potential difference of 16 V discharges and loses 40% of its energy. What is its new p.d.?

Question 19.3 For an isolated metal sphere of radius 30 cm situated in vacuum, what is the electric potential energy stored when charged to a potential of 120 kV?

^[8] Rigorously speaking, this electrical potential energy is stored within the *electric fields* between metal plates of capacitor.

19.3 capacitor networks

19.3.1 capacitors in parallel

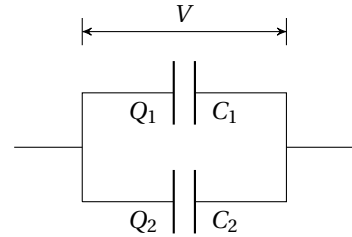
consider two capacitors connected in parallel

same p.d. V across the network: $V = V_1 = V_2$

but charge Q is shared: $Q_{\text{total}} = Q_1 + Q_2$

$$\frac{Q_{\text{total}}}{V} = \frac{Q_1}{V} + \frac{Q_2}{V}$$

$$C_{\text{total}} = C_1 + C_2$$



if three or more capacitors in parallel: $C_{\text{total}} = C_1 + C_2 + C_3 + \dots$

$$Q_{\text{total}} = Q_1 + Q_2 + Q_3 + \dots$$

➤ adding extra capacitor in parallel to a network, total capacitance will increase

explanation: when several capacitors connected in parallel, equivalent to a single capacitor with larger plates, so more charge on the plates $\Rightarrow C \uparrow$

Example 19.4 A capacitor with capacitance C_0 is charged to a p.d. V_0 . It is disconnected from the power supply, and then connected across an identical capacitor. Discuss the change in p.d., and change in energy stored in the system.

✎ initial charge $Q = C_0 V_0$, initial energy stored $W_0 = \frac{1}{2} C V_0^2$

combined capacitance: $C = C_0 + C_0 = 2C_0$

charge is conserved, so final p.d. across: $V = \frac{Q}{C} = \frac{C_0 V_0}{2C_0} = \frac{1}{2} V_0$

charge shared between capacitors, the first capacitor loses half its charge and p.d.

final energy stored in system: $W = \frac{1}{2} C V^2 = \frac{1}{2} \times 2C_0 \times \left(\frac{1}{2} V_0\right)^2 = \frac{1}{4} C_0 V_0^2 \Rightarrow W = \frac{1}{2} W_0$

half of stored energy is lost as heat when electrons flow between two capacitors □

Question 19.4 A capacitor A of capacitance C and a second capacitor B of capacitance $3C$ are connected in parallel. If a voltage is applied across the network, what is the ratio of energy stored in A to that in B ? What about the ratio of electric charge?

19.3.2 capacitors in series

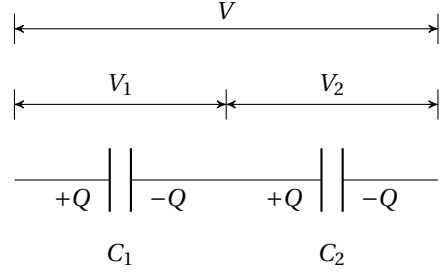
consider next two series capacitors

same charge Q on each plate: $Q = Q_1 = Q_2$ ^[9]

p.d. is shared: $V_{\text{total}} = V_1 + V_2$

$$\frac{V_{\text{total}}}{Q} = \frac{V_1}{Q} + \frac{V_2}{Q}$$

$$\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2}$$



if three or more capacitors in series:

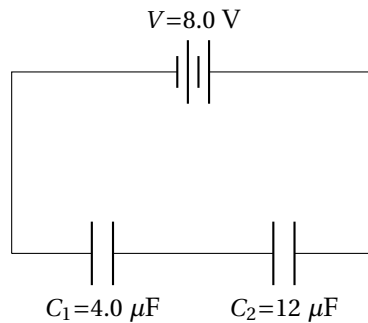
$$\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

$$V_{\text{total}} = V_1 + V_2 + V_3 + \dots$$

➤ adding extra capacitor in series to a network, total capacitance will decrease

explanation: when several capacitors connected in series, equivalent to a parallel-plate capacitor with greater separation, so more charge on the plates $\Rightarrow C \searrow$

Example 19.5 For the circuit shown below, find the p.d. across each of the capacitor.



🔧 using result for combined capacitance, we have: $C_{\text{total}} = \left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1} = \left(\frac{1}{4.0} + \frac{1}{12} \right)^{-1} = 3.0 \mu\text{F}$

charge for the network: $Q = C_{\text{total}} V_{\text{total}} = 3.0 \times 8.0 = 24 \mu\text{C}$

series network so all capacitors have same Q , so: $Q_1 = Q_2 = 24 \mu\text{C}$

p.d. across each individual capacitor: $V_1 = \frac{Q_1}{C_1} = \frac{24}{4.0} = 6.0 \text{ V}$, $V_2 = \frac{Q_2}{C_2} = \frac{24}{12} = 2.0 \text{ V}$

alternatively, we can use properties of series network: $Q_1 = Q_2$ and $V_{\text{total}} = V_1 + V_2$

we can solve simultaneous equations:
$$\begin{cases} 4.0V_1 = 12V_2 \\ V_1 + V_2 = 8.0 \end{cases} \Rightarrow \begin{cases} V_1 = 6.0 \text{ V} \\ V_2 = 2.0 \text{ V} \end{cases}$$

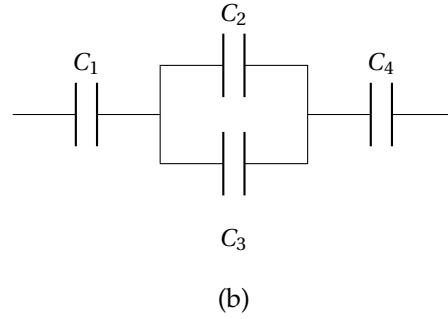
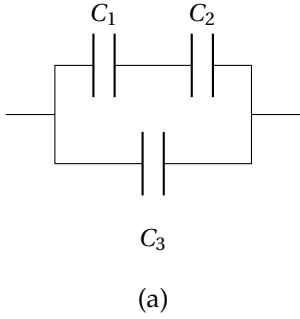
□

^[9]Initially, the H-shaped isolated section between capacitors is uncharged. Since no charge can enter or leave this section, its net charge must remain zero, so same charge Q on each plate of series capacitors.

19.3.3 capacitor networks

more complicated capacitor networks can be considered as a combination of some smaller networks with capacitors in parallel or in series

Example 19.6 $C_1 = C_2 = C_3 = C_4 = 10 \mu\text{F}$, calculate the capacitance of the network (a) and (b).



$$(a) \quad C_{12} = \left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1} = \left(\frac{1}{10} + \frac{1}{10} \right)^{-1} = 5.0 \mu\text{F}$$

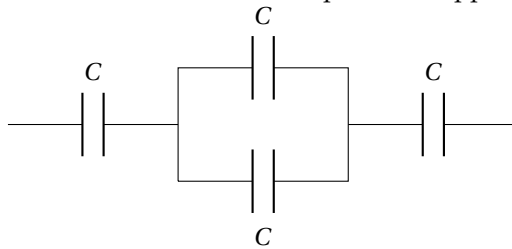
$$C_{\text{total}} = C_{12} + C_3 = 5 + 10 = 15 \mu\text{F}$$

$$(b) \quad C_{23} = C_2 + C_3 = 10 + 10 = 20 \mu\text{F}$$

$$C_{\text{total}} = \left(\frac{1}{C_1} + \frac{1}{C_{23}} + \frac{1}{C_4} \right)^{-1} = \left(\frac{1}{10} + \frac{1}{20} + \frac{1}{10} \right)^{-1} = 4.0 \mu\text{F}$$

□

Example 19.7 Four identical capacitors are arranged as shown. Each capacitor can withstand a maximum p.d. of 12 V, what is the maximum safe p.d. to be applied between the terminals?



suppose charge on the leftmost capacitor is Q , then charge of rightmost capacitor is also Q

but for the two capacitors in parallel, charge is shared, so each has charge $\frac{Q}{2}$

p.d. across the parallel network is half of the p.d. across the two capacitors near the ends

so maximum p.d. across the terminals: $V_{\text{max}} = 12 + 6 + 12 = 30 \text{ V}$

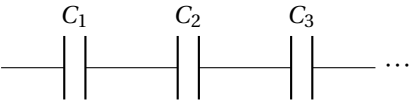
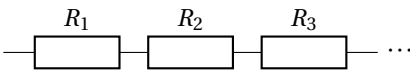
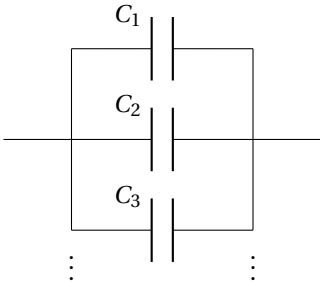
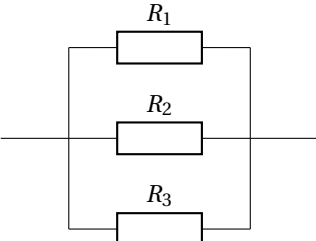
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Question 19.5 Using at most four capacitors of $24 \mu\text{F}$, design a network that has a combined capacitance of (a) $72 \mu\text{F}$, (b) $8 \mu\text{F}$, (c) $36 \mu\text{F}$, and (d) $18 \mu\text{F}$.

Question 19.6 If the two networks in Example 19.6 are both connected to a supply voltage of 15 V, determine the p.d. across each individual capacitor.

19.3.4 capacitors & resistors

as a quick review, we compare capacitor networks with resistor networks

	capacitors	resistors
in series	same charge  $\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots$	same current  $R_{\text{total}} = R_1 + R_2 + R_3 + \cdots$
in parallel	same p.d. across  $C_{\text{total}} = C_1 + C_2 + C_3 + \cdots$	same p.d. across  $\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots$

19.4 charging & discharging capacitors

in this section, we will investigate how the p.d. across a capacitor changes with time when it is being charged, and we will also look into discharging processes

19.4.1 charging phase

initial state: no charge in capacitor: $Q(0) = 0, V_C(0) = 0$

at any instant: $dQ = C dV_C$

current in circuit: $I = \frac{V_R}{R} = \frac{\mathcal{E} - V_C}{R}$

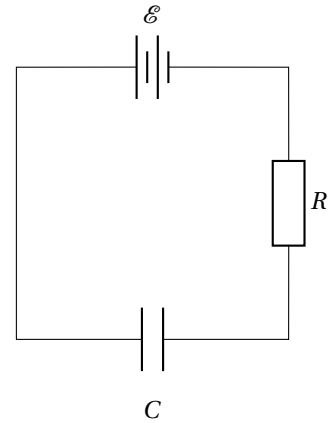
change of charge: $dQ = I dt = \frac{\mathcal{E} - V_C}{R} dt = C dV_C$

$$\frac{dt}{RC} = \frac{dV_C}{\mathcal{E} - V_C}$$

$$\int_0^t \frac{dt}{RC} = \int_0^{V_C} \frac{dV_C}{\mathcal{E} - V_C}$$

$$\frac{t}{RC} \Big|_0^t = -\ln(\mathcal{E} - V_C) \Big|_0^{V_C}$$

simplify everything, we get: $V_C = \mathcal{E} (1 - e^{-t/RC})$



➤ p.d. of capacitor increases at a decreasing rate when it is being charged

as electric charges are separated onto the two plates, pushing more $+Q/-Q$ onto $+ve/-ve$ plate requires more work done to overcome the repulsion $\Rightarrow$ increase in p.d. slows down

➤ p.d of capacitor eventually tends to the battery e.m.f.

charge will continue to flow if there exists a potential difference

when $V_C = \mathcal{E}$, no charge flow, hence charging current gradually drops to zero

19.4.2 discharging phase

capacitor initially charged with $Q(0) = Q, V_C(0) = V_0$

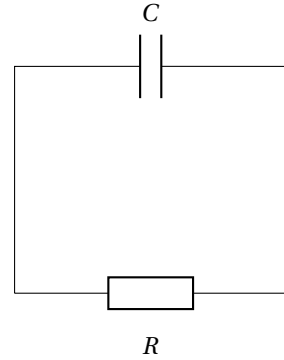
at any instant, $dQ = -C dV_C$ (minus sign because charge decreases during discharging)

also $V_C = V_R$ because C and R are in parallel

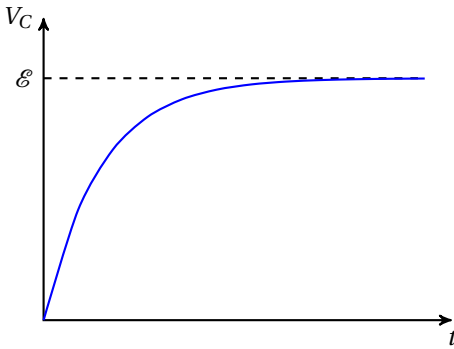
charge change: $dQ = Idt = \frac{V_C}{R} dt = -CdV_C$

$$\begin{aligned} -\frac{dt}{RC} &= \frac{dV_C}{V_C} \\ -\int_0^t \frac{dt}{RC} &= \int_{V_0}^{V_C} \frac{dV_C}{V_C} \\ -\frac{t}{RC} \Big|_0^t &= \ln(V_C) \Big|_{V_0}^{V_C} \end{aligned}$$

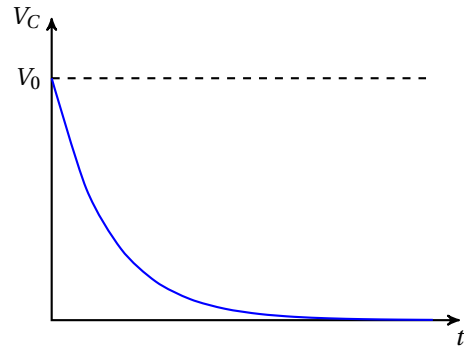
simplify everything, we get: $V_C = V_0 e^{-t/RC}$



- p.d. of capacitor gradually drops to zero during discharging
- discharging current also gradually approaches zero



charging phase of capacitor



discharging phase of capacitor

19.4.3 time constant

$\tau \equiv RC$ is called **time constant**, which determines charging and discharging rate of a capacitor

$R \nearrow \Rightarrow$ smaller charging/discharging current $\Rightarrow$ takes longer to charge/discharge

$C \nearrow \Rightarrow$ more charge to be charged/discharged $\Rightarrow$ takes longer

charging or discharging of capacitors is not instantaneous, there is always a certain delay

as a rule of thumb, after a time $t = 3 \sim 5\tau$, charging or discharging is almost complete